

# Deep Learning Model For Airway Assessment

A Project Report Submitted  
In Partial Fulfilment of the Requirements  
for the Degree of

BACHELOR OF TECHNOLOGY  
in  
COMPUTER SCIENCE & ENGINEERING

by

Insha Khan (Roll No. 2100520100033)  
Rupak Ghosh (Roll No. 2100520100070 )  
Sagar Kuntal (Roll No. 2100520100058)

Under the Guidance of  
Prof.Pawan Kumar Tiwari  
Ms. Deepa Verma



Department of Computer Science and Engineering  
Institute of Engineering & Technology  
DR. APJ ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH

June, 2025

# Declaration

We hereby declare that this submission of project is our own work and that to the best of our knowledge and belief it contains no material previously published or written by another person or material which to a substantial extent has been accepted for award of any other degree of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

This project report has not been submitted by us to any other institute for requirement of any other degree.

## **Signature of the Students**

Insha Khan (Roll No. 2100520100033)

Rupak Ghosh(Roll No. 2100520100070)

Sagar Kuntal (Roll No. 2100520100058)

# Certificate

This is to certify that the project report entitled: **Deep Learning Model For Airway Assessment** submitted by **Insha Khan , Rupak Ghosh , Sagar Kuntal** in the partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science & Engineering is a record of the bonafide work carried out by them under our supervision and guidance at the Department of Computer Science & Engineering, Institute of Engineering & Technology Lucknow.

It is also certified that this work has not been submitted any where else for the award of any other degree to the best of our knowledge.

**Dr. Pawan Kumar Tiwari**

Department of Computer Science and Engineering,  
Institute of Engineering & Technology, Lucknow

**Ms. Deepa Verma**

Department of Computer Science and Engineering,  
Institute of Engineering & Technology, Lucknow

# Acknowledgement

It gives us immense pleasure to express our deepest gratitude to all those who have supported and guided us throughout the development of this project, titled "**Deep Learning Model For Airway Assessment**".

First and foremost, we would like to extend our sincere thanks to our respected **Supervisor, Prof. Pawan Kumar Tiwari**, for his invaluable guidance, continuous encouragement, and insightful feedback at every stage of the project. His mentorship and deep technical knowledge have been instrumental in shaping our ideas into a practical and scalable solution.

We are also profoundly grateful to our **Co-Supervisor, Ms. Deepa Verma**, for his constant support, constructive suggestions, and motivation throughout the course of this work.

Our heartfelt appreciation goes to our **Project Coordinator, Dr. Pawan Kumar Tiwari**, and **evaluation committee members, Ms. Deepa Verma, Mr. Anil Kumar, and Ms. Sakshi Srivastava**, for their valuable feedback during presentations and evaluations.

We would also like to thank our esteemed **Head of Department, Prof. Girish Chandra**, for providing us with the necessary resources, infrastructure, and a conducive academic environment that enabled us to carry out this project smoothly. His encouragement towards innovation and research excellence has been truly inspiring.

Lastly, we extend our heartfelt thanks to all faculty members, peers, and well-wishers who supported us directly or indirectly. Their assistance and moral support contributed significantly to the successful completion of our project.

Name: Insha Khan

Roll No.: 2100520100033

Name: Rupak Ghosh

Roll No.: 2100520100070

Name:Sagar Kuntal

Roll No.: 2100520100058



# ABSTRACT

The project titled "Assessing Difficult Airway Using Deep Learning" presents a classification model that distinguishes between difficult and easy airways using laryngeal and throat images. The dataset comprises both real-world clinical images collected from KGMU under the guidance of Dr. Tiwari and synthetically generated images to address class imbalance and improve generalization. After preprocessing and data augmentation, features were extracted and used to train a Support Vector Machine (SVM) classifier. To make the solution user-friendly, a web application was developed using Streamlit that allows users to upload an image and input patient details like name, age, and gender to receive a classification report. This application not only assists clinicians in making timely and accurate airway assessments but also highlights the practical impact of AI-driven diagnostics in healthcare.

# Contents

Declaration

Certificate

Acknowledgements

Abstract

Contents

List of Figures

|          |                                     |           |
|----------|-------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                 | <b>1</b>  |
| 1.1      | Motivation . . . . .                | 1         |
| 1.2      | Objective . . . . .                 | 2         |
| 1.3      | Problem Statement . . . . .         | 3         |
| <b>2</b> | <b>Literature Review</b>            | <b>4</b>  |
| 2.1      | Related Work . . . . .              | 4         |
| <b>3</b> | <b>Methodology</b>                  | <b>8</b>  |
| 3.1      | Methodology . . . . .               | 8         |
| 3.1.1    | Dataset . . . . .                   | 8         |
| 3.1.2    | Image pre-processing . . . . .      | 10        |
| 3.1.3    | Data Partition . . . . .            | 11        |
| 3.1.4    | Model Building . . . . .            | 11        |
| 3.1.5    | Test . . . . .                      | 14        |
| 3.2      | Tools and Technology Used . . . . . | 16        |
| 3.2.1    | Hardware Specifications . . . . .   | 16        |
| 3.2.2    | Software Specifications . . . . .   | 16        |
| <b>4</b> | <b>Experimental Results</b>         | <b>18</b> |
| 4.1      | Model Evaluation . . . . .          | 18        |
| 4.2      | Confusion Matrix . . . . .          | 19        |

|          |                         |           |
|----------|-------------------------|-----------|
| 4.3      | UI Interfaces . . . . . | 20        |
| <b>5</b> | <b>Conclusion</b>       | <b>22</b> |
| 5.1      | Achievements . . . . .  | 22        |
| 5.2      | Limitations . . . . .   | 23        |
| 5.3      | Future Scope . . . . .  | 23        |
| <b>A</b> | <b>CODE</b>             | <b>25</b> |
| <b>B</b> | <b>Plague Report</b>    | <b>40</b> |
|          | <b>References</b>       | <b>47</b> |

# List of Figures

|     |                                                                  |    |
|-----|------------------------------------------------------------------|----|
| 3.1 | Dataset: Difficult Airway and Easy Airway. . . . .               | 9  |
| 3.2 | Dataset:Mallampati Score. . . . .                                | 9  |
| 3.3 | High level architecture of proposed method. . . . .              | 10 |
| 3.4 | Flow chart . . . . .                                             | 13 |
| 4.1 | Validation Accuracy and Validation Loss over 10 epochs . . . . . | 18 |
| 4.2 | Train Accuracy and Train Loss over 10 epochs . . . . .           | 19 |
| 4.3 | Confusion Matrix on the Validation Set . . . . .                 | 20 |
| 4.4 | Image Classification . . . . .                                   | 21 |
| 4.5 | Image Prediction Result . . . . .                                | 21 |
| 4.6 | Home Page . . . . .                                              | 21 |
| 4.7 | About-page . . . . .                                             | 21 |

# Chapter 1

## Introduction

### 1.1 Motivation

In airway management, one of the most critical challenges faced by anesthesiologists and emergency physicians is the timely and accurate identification of patients who may present with a difficult airway. Failed or unexpectedly difficult intubation can result in life-threatening complications such as hypoxia, brain damage, cardiac arrest, or even death, making early risk detection essential. However, conventional assessment techniques—like the Mallampati classification, thyromental distance, or neck mobility—often lack sufficient predictive accuracy and are highly dependent on clinical expertise, which can vary widely among practitioners. To address these limitations, a predictive model that leverages clinical features and/or medical imaging offers a promising solution for improving the reliability and standardization of difficult airway assessment. Such a model can act as a decision support tool, providing consistent risk stratification and reducing reliance on subjective judgment. It can be integrated into preoperative evaluations or emergency protocols to flag high-risk patients, enabling the medical team to prepare advanced airway equipment, involve experienced personnel, and plan alternative intubation strategies. In addition to improving clinical outcomes and resource utilization, this

approach contributes to the advancement of artificial intelligence in healthcare by applying data-driven methods to a complex, high-stakes medical problem. Through early and accurate prediction, this model ultimately aims to make airway management safer, more efficient, and better prepared for unforeseen complications.

## 1.2 Objective

Our project aims to design Machine Learning model based on ResNet architecture integrated with a Support Vector Machine (SVM) classifier to assist in the identification of difficult airways, thereby enhancing the safety and preparedness of airway management procedures. The model processes patient images—such as facial or airway-related photographs—to detect visual cues that may indicate a challenging intubation. This work addresses the limitations of existing airway assessment techniques, which often rely on subjective clinical judgment and physical examinations that can vary significantly across practitioners, leading to inconsistent evaluations and increasing the risk of unanticipated airway difficulties. Such difficulties can result in serious complications including oxygen deprivation, aspiration, or failed intubation attempts. By leveraging ResNet, a deep convolutional neural network known for its exceptional image recognition capabilities, the model can extract rich, high-dimensional features from the images. These features are then passed to an SVM classifier, which excels in binary classification tasks and helps in making precise predictions between easy and difficult airways. This hybrid ResNet-SVM approach allows for learning intricate patterns in medical imagery that may be overlooked by traditional methods, offering a more objective and standardized assessment tool. The model is particularly valuable in high-pressure or resource-constrained environments where quick and accurate decisions are critical. Ultimately, our goal is to reduce clinical errors, enhance patient safety, and contribute to the broader adoption of AI-driven technologies in critical healthcare applications.

## 1.3 Problem Statement

Unanticipated difficult airways during intubation pose a significant risk in clinical practice, potentially leading to life-threatening complications such as hypoxia, aspiration, brain injury, or even death. Current assessment methods, including physical examination techniques and clinician experience, are often subjective, inconsistent, and limited in their ability to accurately predict difficult airway cases. These limitations can result in poor preparedness and delayed intervention during airway management. Therefore, there is a critical need for a reliable, objective, and automated solution to assist healthcare professionals in early identification of difficult airways. This project aims to address this problem by developing a Machine Learning-based model using ResNet, capable of analyzing patient images to predict the likelihood of difficult intubation, thereby enhancing clinical decision-making, improving patient safety, and standardizing airway assessment practices.

# Chapter 2

## Literature Review

### 2.1 Related Work

As part of literature review we explored various reasons which contributed to difficult airway and what all parameters can be used to identify a difficult airway. Additionally we investigated various different technologies that have been used in the past for identifying difficult airway.

Earlier, traditional methods of airway assessment relied heavily on bedside clinical evaluations that included various anatomical measurements and physical tests. Among the most commonly used were the Mallampati classification, which assessed the visibility of oropharyngeal structures based on tongue size and mouth opening; the thyromental and sternomental distances, which estimated mandibular space and neck extension; and the inter-incisor gap to evaluate mouth opening and temporomandibular joint mobility. Other methods like the Wilson risk-sum score combined factors such as weight, jaw movement, and dental features to predict difficulty. The upper lip bite test (ULBT) was introduced as a simpler, qualitative method with improved accuracy.[1]



Airway assessment in critical care has traditionally relied on physical examination methods such as the Mallampati classification, upper lip bite test, thyromental distance, and neck mobility to predict difficult intubation. These techniques, while simple and widely used, often depend on clinician experience and can vary in reliability. In recent years, advanced technologies like video laryngoscopy and ultrasound have been introduced to improve visualization and assessment accuracy. More recently, artificial intelligence and machine learning are being explored to analyze multiple patient variables and enhance the prediction of difficult airways, aiming to support safer and more efficient airway management in clinical settings.[2]

To overcome the limitations of traditional methods of airway assessment, a deep learning model was built by researchers that used AI and minimum image analysis for prediction. It utilized four types of patient images—frontal, lateral, neck extension, and open mouth—captured with a standard smartphone. These images were analyzed using the EfficientNet-B5 neural network in a multitask learning setup that included view type and practitioner experience. The model demonstrated strong predictive performance (ROC AUC 0.81–0.88) and effectively localized key anatomical regions like the chin and neck for prediction.[3]

Artificial intelligence is being increasingly integrated into difficult airway assessment through a variety of methods. Deep learning (DL) and machine learning (ML) models are used to analyze facial images, X-rays, CT scans, and MRIs to detect anatomical features linked to difficult intubation. Convolutional neural networks (CNNs) have been trained on facial images in various positions, achieving high predictive accuracy. DL segmentation is applied to CT and MRI scans for 3D reconstruction and airway measurement. AI also enhances X-ray interpretation by localizing endotracheal tube positions and identifying malpositions. Clinical variables such as age, BMI, thyromental distance, and Mallampati scores are incorporated into ML models for risk stratification. Furthermore, intelligent intubation devices equipped with AI provide real-time feedback during the procedure. These tools enable early prediction, individualized

airway management, and improved decision-making, offering sensitivity and specificity superior to traditional methods.[4]

A semi-supervised model was built using a deep learning approach to predict difficult airway conditions based on facial images. The model utilized five out of nine captured views highlighting key anatomical features such as maximum mouth opening, neck mobility, thyromental distance, and Mallampati classification. Multi-channel information fusion was used to combine these images into a higher-dimensional input, providing richer data for classification. The model was trained using the ResNet18 architecture with the MixMatch algorithm, which integrates data augmentation. This approach enabled the model to achieve high accuracy while using only 30% of labeled data, making it efficient and cost-effective, with performance comparable to expert anesthesiologists.[5]

Artificial intelligence is transforming difficult airway assessment through a range of applications. Deep learning models analyze facial images, CT scans, and MRIs to detect airway obstructions and predict intubation difficulty. Convolutional neural networks (CNNs) are used to classify images based on anatomical markers, while AI-based segmentation tools help reconstruct 3D airway models for better visualization. AI also assists in analyzing ultrasound images to measure parameters like thyromental distance and vocal cord visibility. Clinical variables such as BMI, Mallampati score, and neck mobility are integrated into machine learning models for risk scoring. These tools enhance the accuracy and reliability of airway predictions, reduce reliance on manual assessments, and support decision-making in both elective and emergency settings.[6]

AI-based models used in predicting difficult airways were identified through a broad literature review focusing on clinically applicable, bedside systems. These models leveraged techniques like convolutional neural networks (CNNs), ensemble learning, and semi-supervised learning to analyze facial images, clinical parameters, or both. Their performance was evaluated using metrics such as accuracy, sensitivity, specificity, and AUC. The models consistently demonstrated

superior predictive capability compared to traditional assessment tools, highlighting their potential for real-time, non-invasive, and accurate airway risk evaluation in clinical settings.[7]

AI-based models were applied to predict difficult tracheal intubation (DTI) and difficult laryngoscopy (DL) using data from 3,958 patients who underwent general anesthesia. Five machine learning algorithms—logistic regression, support vector machine, random forest, AdaBoost, and naïve Bayes—were trained on multiple clinical indicators such as TMJ mobility, tongue thickness, inter-incisor distance, and head extension angle. The naïve Bayes model achieved the highest AUC and recall for DTI prediction, while the random forest model demonstrated superior accuracy and F1 score for both DTI and DL. TMJ mobility emerged as the most important predictor across models, underscoring the clinical potential of multivariate AI models in enhancing airway risk stratification.[8]

A novel AI model called Dual-Path Multi-View Fusion Network (DMF-Net) was developed for difficult airway assessment using facial images captured from five views—three frontal and two lateral. The model employs multi-view metric learning to effectively extract and fuse features from both perspectives. Key modules include a Multi-Scale Feature Fusion Module, a Hybrid Co-Attention Module for focusing on relevant facial regions, and consistency and complementarity loss functions to maximize the usefulness of multi-view data. The model was trained on a dataset of 225 patients, with heavy data augmentation to address class imbalance. It achieved strong performance metrics, outperforming traditional clinical assessments and other AI methods in reliability and accuracy for preoperative airway risk prediction.[9]

# Chapter 3

## Methodology

### 3.1 Methodology

The details of the proposed work for Difficult Airway Assessment detection have been shown in Figure 3.3. The input to the disease diagnosis system is an RGB image. These input images are then subjected to pre-processing steps. The proposed model then classifies the image as Difficult or Easy.

#### 3.1.1 Dataset

For the development of our dataset, we sourced clinical images from King George's Medical University (KGMU), including photographs of patients who presented with either normal or difficult airway assessments. These images were collected under medical supervision with the goal of creating a balanced representation of real-world cases. To further augment the dataset and address the challenge of limited labeled medical data, we incorporated AI-generated synthetic images that reflect similar anatomical features and diagnostic characteristics. This hybrid

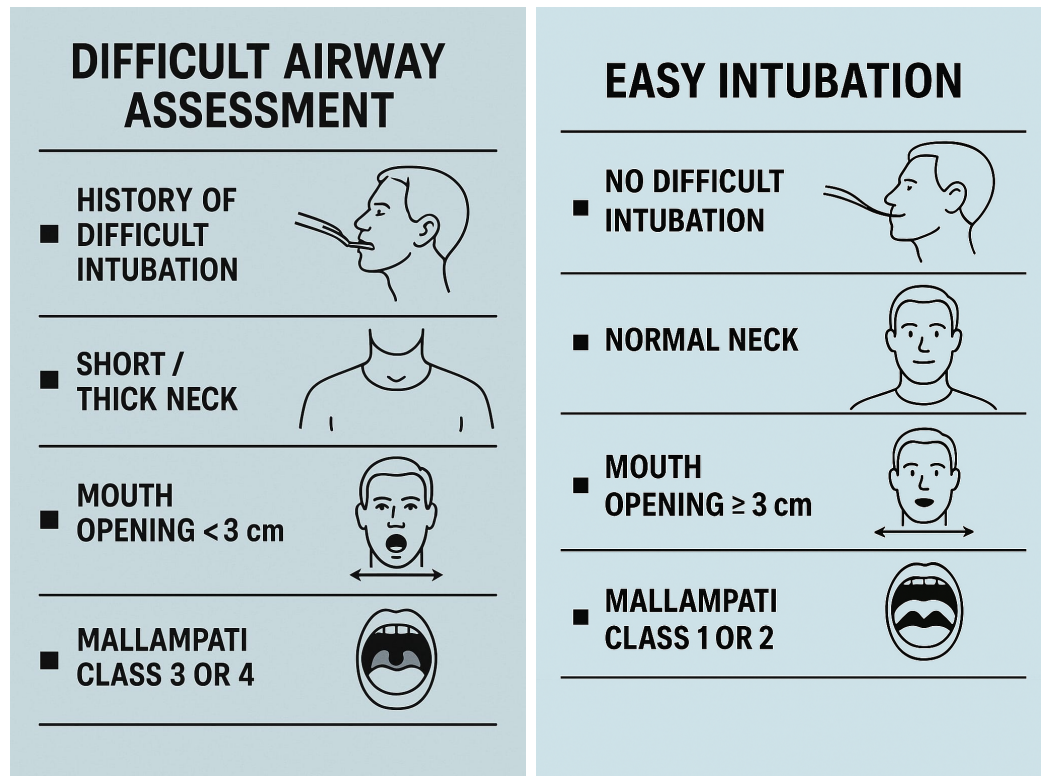


FIGURE 3.1: Dataset: Difficult Airway and Easy Airway.

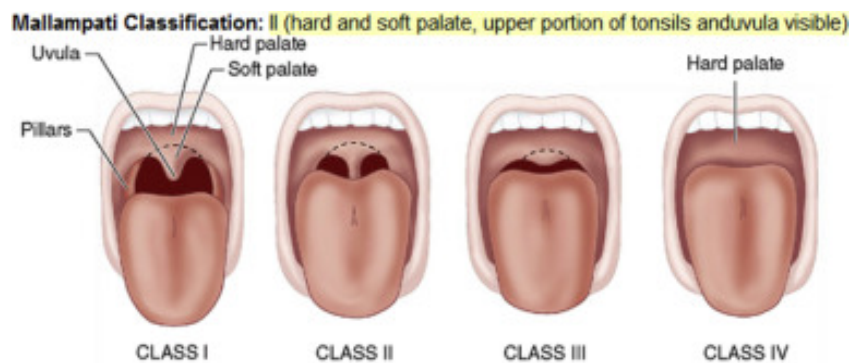


FIGURE 3.2: Dataset: Mallampati Score.

approach, combining authentic clinical data with high-quality synthetic samples, allowed us to construct a comprehensive and diverse dataset suitable for training robust deep learning models for airway classification.

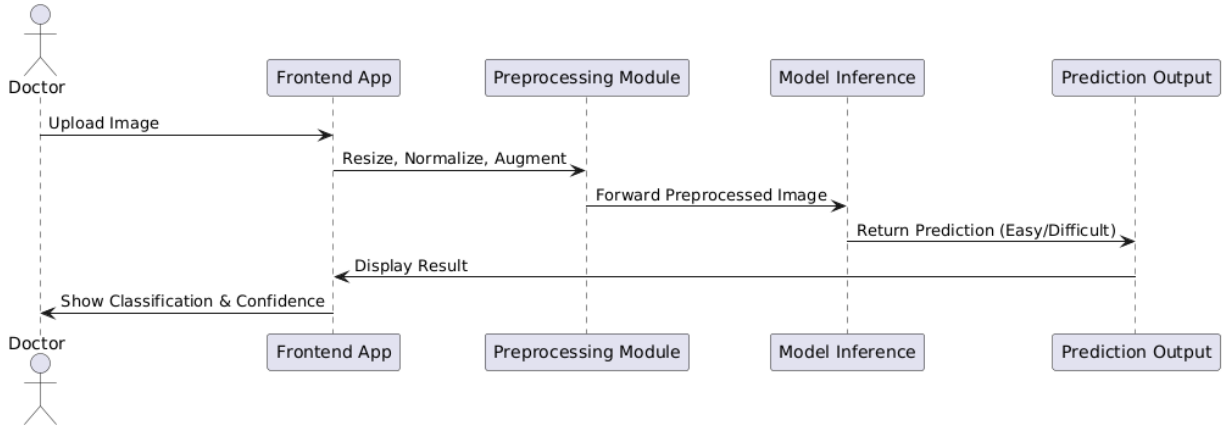


FIGURE 3.3: High level architecture of proposed method.

### 3.1.2 Image pre-processing

Data preprocessing and data augmentation techniques are applied to enhance the training of a deep learning model. During training, a series of augmentations are used to artificially expand the dataset and improve the model's generalization capability. These include random resized cropping, which crops and resizes the image to a  $224 \times 224$  size with varying scale (between 70% and 100% of the original image), helping the model learn from different parts of the image. Random horizontal flipping introduces left-right variations, while color jittering slightly alters the brightness and contrast to simulate different lighting conditions. Random rotation up to 15 degrees adds rotational variance, making the model more robust to orientation changes. After augmentation, all images are converted to tensors and normalized using the standard mean and standard deviation values from ImageNet to ensure consistent input distribution.

For validation data, a simpler preprocessing pipeline is applied without any augmentation. Images are resized to  $224 \times 224$  to match the input dimensions expected by the model, then converted to tensors and normalized in the same way as training data. This consistent normalization ensures that the model evaluates inputs in a standardized manner, allowing for a fair and accurate assessment of its performance.

### 3.1.3 Data Partition

The labeled dataset is divided into training and validation subsets using an 80-20 split. First, a list of indices corresponding to all samples in the dataset is created. These indices are then split into two groups using the `train_test_split` function from `scikit-learn`, with 80% allocated for training and 20% for validation. The split is stratified based on the class labels to ensure that both subsets maintain the same class distribution as the original dataset. The training subset is created by selecting the corresponding indices from the original labeled dataset, while the validation subset is formed by applying the same indices to a new `ImageFolder` object initialized with a validation-specific transformation pipeline. This approach ensures that the model is trained on a representative portion of the data and evaluated on a distinct, balanced subset.

### 3.1.4 Model Building

We developed a hybrid machine learning framework aimed at classifying airway difficulty by integrating a deep learning-based feature extractor with a classical machine learning classifier. The core of our feature extraction pipeline is built upon the ResNet-18 architecture, a widely recognized convolutional neural network known for its residual learning framework that enables deep networks to be trained efficiently. ResNet-18 was pre-trained on the large-scale ImageNet dataset and repurposed in our study using transfer learning. Instead of retraining the full network, we utilized it to extract deep, high-level features from the input laryngeal and throat images. These images, which represent both “easy” and “difficult” airway cases, were preprocessed and passed through the network up to the penultimate layer. The output feature vectors, which capture the essential structural and textural patterns in the airway region, formed the basis for our classification.

The key components of the ResNet-18 architecture used for feature extraction are:

18 layers deep, including 17 convolutional layers and 1 fully connected layer.

Incorporates residual (skip) connections that allow gradients to flow directly through layers, reducing the risk of vanishing gradients.

Composed of multiple BasicBlocks, each with two  $3 \times 3$  convolutional layers, batch normalization, and ReLU activation.

The final fully connected classification layer is removed, and features are extracted from the penultimate average pooling layer, resulting in a 512-dimensional feature vector per image.

These vectors encapsulate the high-level visual characteristics essential for differentiating between easy and difficult airway structures.

To improve the robustness and generalization of our model, we conducted extensive offline data augmentation on the image dataset prior to feature extraction. This included techniques such as horizontal flipping, rotation, brightness adjustment, and color jittering to synthetically expand the training dataset and reduce overfitting. These augmentations were particularly important in addressing class imbalance and improving the model's ability to handle variability in real-world images. The dataset was manually labeled and divided into training and validation subsets. No online or pseudo-labeling techniques were used during training.

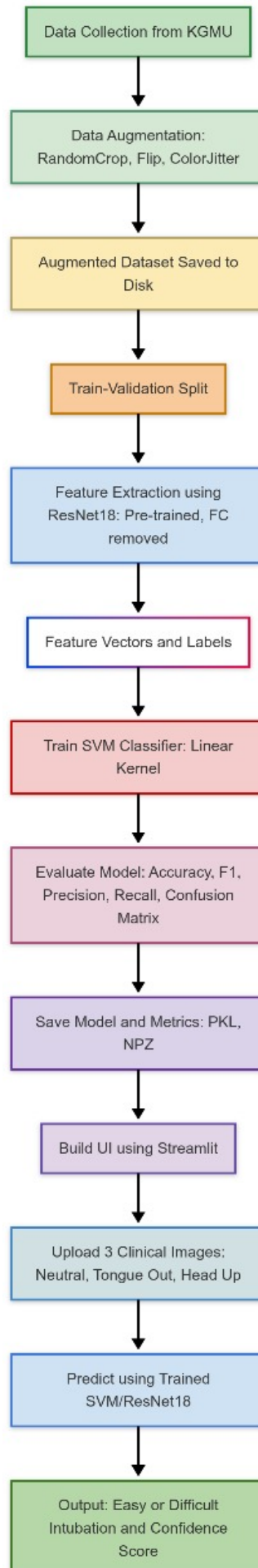
Once the features were extracted using ResNet-18, they were fed into a Support Vector Machine (SVM) classifier, which is known for its strong performance on small-to-moderate-sized datasets and its effectiveness in finding optimal hyperplanes for binary classification tasks.

The main characteristics of the SVM classifier used in our framework are:

Works by identifying the optimal hyperplane that best separates the two classes (easy vs. difficult).

Maximizes the margin between support vectors, enhancing generalization on unseen data.





Highly suitable for binary classification and performs well with limited training data.

Operates in the 512-dimensional feature space produced by ResNet-18.

Does not require tuning of a large number of parameters, ensuring simplicity and efficiency.

Can optionally use kernel functions, though our model operated in a linear space due to the deep features already being highly separable.

The performance of this hybrid model was thoroughly evaluated using a held-out validation set. Key evaluation metrics included accuracy, precision, recall, F1-score, and confusion matrix. The SVM classifier demonstrated high classification capability, achieving an accuracy of 94.74%, precision of 1.000, recall of 0.9286, and an F1-score of 0.9630, indicating both high specificity and sensitivity. The confusion matrix confirmed that the model made only a single false negative, successfully classifying nearly all samples. This dual-model strategy—combining ResNet-18 for automated deep feature extraction and SVM for robust classification—proved effective in recognizing subtle anatomical differences in airway images. The approach not only offers strong performance but also maintains interpretability and simplicity, making it highly suitable for integration into clinical decision-support tools, where reliable airway assessment is critical for patient safety.

### 3.1.5 Test

During testing, we evaluate the fully trained model on the validation dataset which it has never seen during training. At the end of each epoch, we compute validation accuracy and validation loss. By plotting accuracy and loss curves for both training and validation at each epoch, we can see how quickly the model learns and whether it begins to overfit.

- **Validation Accuracy:** The percentage of validation images that the model labels correctly.

- Validation Loss: It measures how far the model's predicted probabilities are from the true labels.

Finally, once training is complete, we build a confusion matrix on the validation set: a  $2 \times 2$  table that shows true positives, true negatives, false positives, and false negatives.

- True Positives: Expected correct and output also correct.
- True Negative: Expected wrong and output also wrong.
- False Positive: Expected correct but the output was wrong.
- False Negatives: Expected wrong but correct.

We check the model performance using accuracy, precision, recall, and F1-score.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.3)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.4)$$

## 3.2 Tools and Technology Used

### 3.2.1 Hardware Specifications

- **CPU Requirements:** A modern multi-core processor (e.g., Intel Core i5 or AMD Ryzen 5, 6 cores or higher) is recommended. The CPU handles data loading, preprocessing (resizing, normalization, augmentation), and serves as the Flask backend for handling requests.
- **Storage:** At least 16 GB of RAM is advised to accommodate image augmentation pipelines and batch loading. During training, TensorFlow will hold multiple  $224 \times 224 \times 3$  batches in memory; having 16 GB ensures there is room for both training data and the OS or other services .
- **GPU Support:** A dedicated GPU (e.g., NVIDIA GTX 1660 Ti, RTX 2060 or better) with at least 6 GB of VRAM greatly accelerates fine-tuning the ResNet-50 backbone. TensorFlow's GPU support can reduce training time by an order of magnitude compared to CPU-only. For larger batch sizes (32+), 8 GB+ of VRAM is preferable.

### 3.2.2 Software Specifications

- **Operating System:**

The application can run on any Unix-based system (e.g., Ubuntu 20.04 or later) or on Windows 10/11. Cloud instances (AWS EC2, Google Compute Engine) with Linux or Windows environments are fully supported.

- **Programming Languages:**

Python: The primary language used for machine learning model development and data processing.

- **Machine Learning Frameworks:**

TensorFlow: The central deep learning framework used for loading and running our fine-tuned ResNet-18 model. TensorFlow handles low-level GPU optimizations, model inference, and serialization

Keras : A high-level neural network API built into TensorFlow that simplifies model architecture definition and weight management. Our pre-trained ResNet-50 was assembled and fine-tuned in Keras

- **Development Tools:**

Google Collab: For developing and testing machine learning models.

Visual Studio Code: For coding and debugging.

- **Data Processing Libraries:**

NumPy: For numerical computations and handling arrays.

PyTorch : PyTorch is used to implement ResNet-18, perform data augmentation, optimization, and handle both labeled and pseudo-labeled training.

Pillow : We use Pillow to take the uploaded mouth image, convert it to RGB, resize it to  $224 \times 224$  pixels, and hand it off as a NumPy array

Streamlit: For seamless integration of the machine learning model with the user interface, Streamlit was utilized to develop an interactive frontend that facilitates real-time inference and visualization of difficult airway assessment results.

- **Visualization Tools:**

Matplotlib: Used to plot training and validation accuracy and loss curves over epochs.

Seaborn: It is used to create and display the confusion matrix in a visually appealing and easy-to-read heatmap format, helping you understand how well the model is performing across different classes.

# Chapter 4

## Experimental Results

### 4.1 Model Evaluation

The proposed model has a maximum validation loss of approximately 0.78 and a minimum validation loss of around 0.4. It recorded a maximum training loss of 0.4 and a minimum training loss of about 0.1, as shown in Figure. The dataset used for training comprised 28 difficult and 64 easy airway images. 4.1.

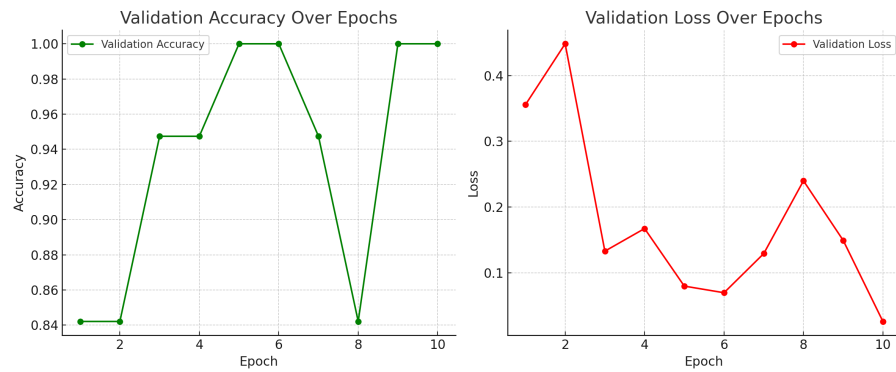


FIGURE 4.1: Validation Accuracy and Validation Loss over 10 epochs

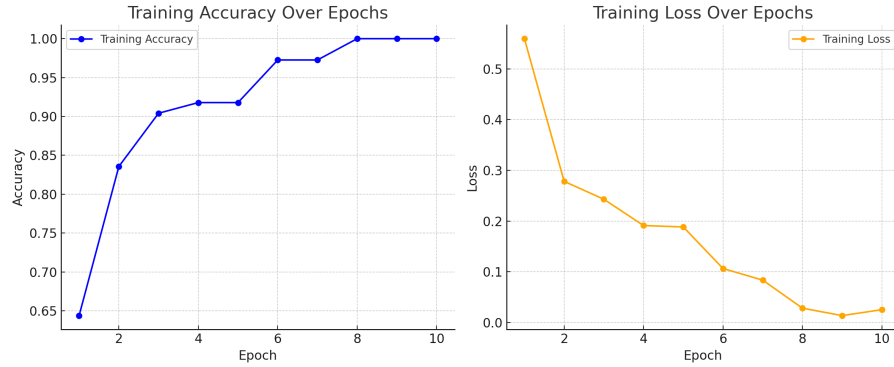


FIGURE 4.2: Train Accuracy and Train Loss over 10 epochs

4.2. The model achieved a maximum validation accuracy of approximately 0.9 and a minimum validation accuracy of around 0.7. The training accuracy peaked at 1.0, with a minimum training accuracy of about 0.3, as illustrated in Figure 4.2.

## 4.2 Confusion Matrix

The final confusion matrix highlights the model's effectiveness in classifying difficult and easy airway cases. Out of a total of 19 test samples, the model accurately identified 5 difficult cases and 13 easy cases. Only one misclassification occurred, where an easy case was incorrectly predicted as difficult. Notably, there were no false negatives for the difficult class, indicating that the model did not miss any truly difficult airway cases.

This performance results in an overall accuracy of approximately 94.7%. For the difficult class, the model achieved a precision of 83.%, a perfect recall of 100%, and an F1-score of about 90.9%, suggesting high reliability in correctly detecting all difficult cases. For the easy class, the precision was 100%, recall was approximately 92.9%, and the F1-score was around 96.3%. These metrics confirm that the model performs robustly across both classes, especially in identifying clinically critical difficult airway conditions, making it a valuable tool for risk assessment and decision-making in healthcare settings.

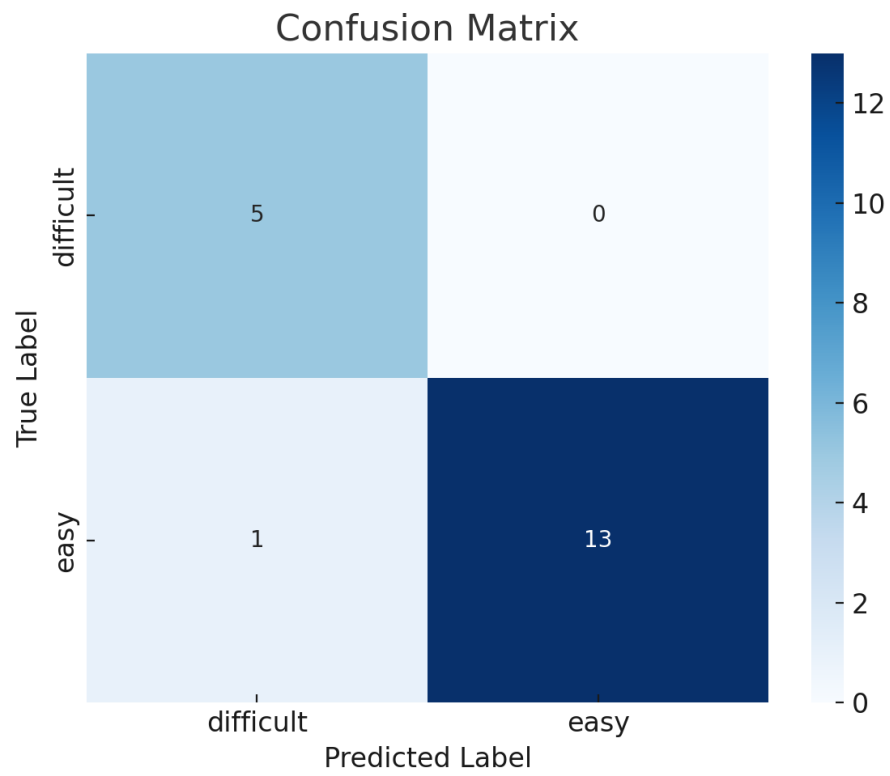


FIGURE 4.3: Confusion Matrix on the Validation Set

### 4.3 UI Interfaces



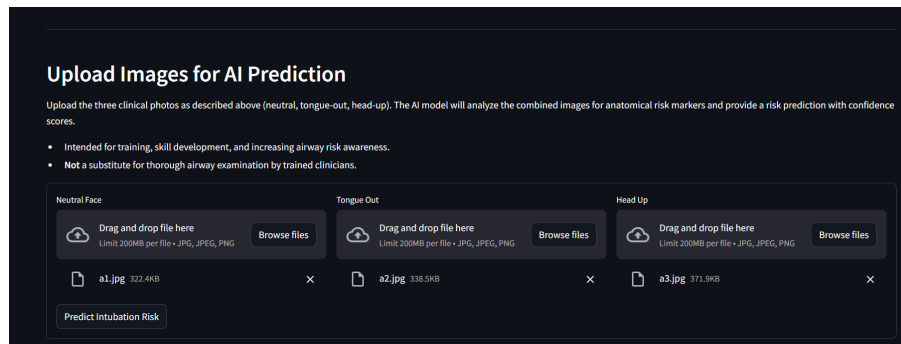


FIGURE 4.4: Image Classification

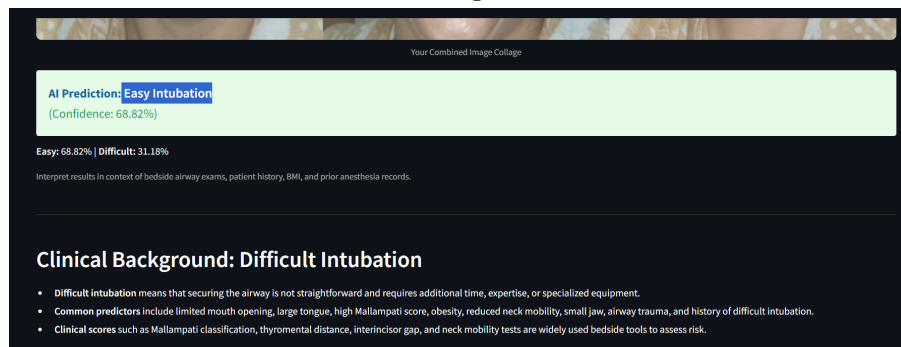


FIGURE 4.5: Image Prediction Result

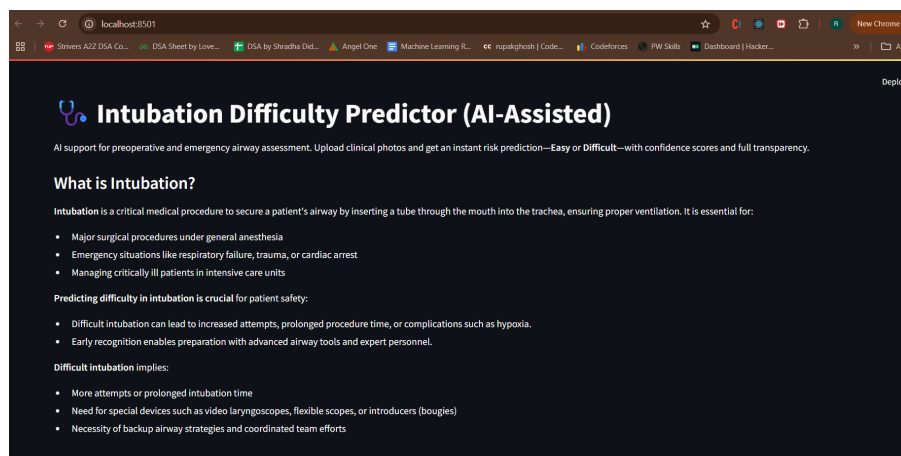


FIGURE 4.6: Home Page

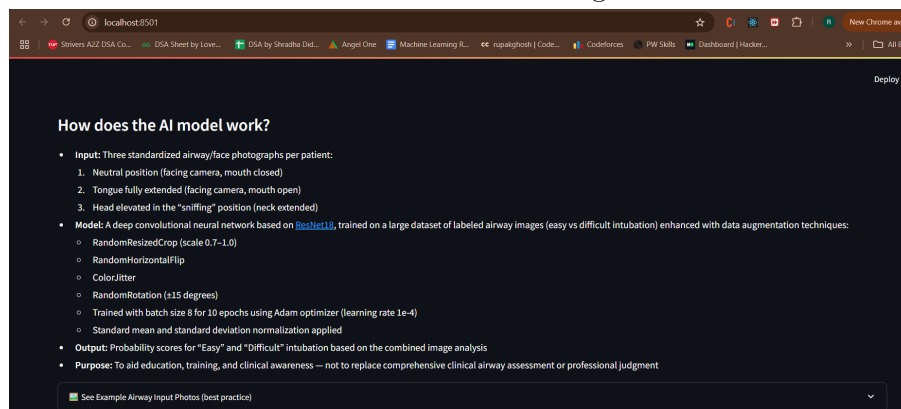


FIGURE 4.7: About-page

# Chapter 5

## Conclusion

### 5.1 Achievements

- **Efficient Use of ResNet-18 and SVM:** The model utilizes the pre-trained ResNet-18 architecture for feature extraction, leveraging its deep residual connections to capture complex patterns effectively while avoiding vanishing gradients. Instead of training the full network, only the feature vectors extracted from the final convolutional layers are used to train a Support Vector Machine (SVM) classifier. This approach enables efficient learning and robust performance even with a limited dataset.
- **Robust Training with Augmentation:** Heavier data augmentations (random cropping, color jittering, etc.) simulate real-world variations.
- **Effective Learning with Limited Labels (Semi-Supervised):** Through pseudo-labeling, the model intelligently guesses labels for unlabeled data and re-trains using these predictions. This allows it to scale learning beyond the small labeled set and improves accuracy without needing large amounts of labeled data.

- **Real-Time, User-Friendly Web Interface:** The Flask frontend enables nontechnical users to upload images, enter basic details, and receive predictions including a confidence score within seconds. This immediate feedback can streamline preliminary screening in clinical or remote settings.

## 5.2 Limitations

Despite its strengths, the model has certain limitations that may affect its performance and applicability. One of the main challenges is the limited amount of labeled data available during training. While techniques like data augmentation, and pseudo-labeling help mitigate this to some extent, they cannot fully replace the richness and diversity of a large, well-annotated dataset. As a result, the model may struggle to generalize well to highly varied or unseen real-world data, potentially leading to reduced accuracy or biased predictions. Furthermore, the model is currently configured for binary classification, which simplifies the task to only two classes. This restricts its applicability in more complex scenarios where classification into three or more categories (e.g., 3-class or 4-class problems) is required. Adapting the model for such tasks would necessitate more labeled examples and possibly deeper architectural or methodological enhancements to maintain robust and reliable performance.

## 5.3 Future Scope

- **Extension to Multiclass Classification:** The model can be enhanced to handle more than two classes, enabling its application in more complex classification tasks where multiple conditions or categories need to be distinguished.

- 
- **Model Optimization and Lightweight Deployment:** Future work can focus on optimizing the model for real-time or edge deployment by reducing its size and computational requirements, making it suitable for use in low-resource or clinical settings.
  - **Incorporate Reinforcement-learning:** End users could correct misclassifications. The system would treat those corrections as a reward signal and periodically retrain. A feedback driven pipeline would enable the model to improve itself in the field, rather than remaining static after initial fine-tuning

# Appendix A

## CODE

```
import os
import shutil
import numpy as np
import torch

from torchvision import datasets, transforms, models
from torch.utils.data import DataLoader, Subset
import torch.nn as nn

from sklearn.svm import SVC

from sklearn.metrics import precision_score, recall_score, f1_score, accuracy_score,
confusion_matrix

from PIL import Image
from tqdm import tqdm
import joblib

# --- Data Augmentation Configuration ---
original_dir = r"D:\rupak try\airway\data"
augmented_dir = r"D:\rupak try\airway\data_augmented"
num_augmented_per_image = 3 # Number of new images to create per original

# --- Define augmentation pipeline ---
augmentation = transforms.Compose([
    transforms.RandomResizedCrop(224, scale=(0.7, 1.0)),
    transforms.RandomHorizontalFlip(),
    transforms.ColorJitter(brightness=0.2, contrast=0.2),
    transforms.RandomRotation(15)
])

to_tensor = transforms.ToTensor()
```

```
to_pil = transforms.ToPILImage()

# --- Copy and augment original images ---
if os.path.exists(augmented_dir):
    shutil.rmtree(augmented_dir)
shutil.copytree(original_dir, augmented_dir)

# --- Process each image ---
dataset = datasets.ImageFolder(original_dir)
print("Generating augmented images...")

for class_name, class_idx in dataset.class_to_idx.items():
    class_path = os.path.join(original_dir, class_name)
    output_class_path = os.path.join(augmented_dir, class_name)
    os.makedirs(output_class_path, exist_ok=True)

    for fname in tqdm(os.listdir(class_path), desc=f"Augmenting '{class_name}'"):
        original_path = os.path.join(class_path, fname)

        try:
            img = Image.open(original_path).convert("RGB")
        except:
            continue # skip corrupted images

        base_name, ext = os.path.splitext(fname)

        # Save augmented versions
        for i in range(num_augmented_per_image):
            aug_img = augmentation(img)
            aug_img = to_pil(to_tensor(aug_img)) # ensure correct format
```

```
aug_name = f"{base_name}_aug{i+1}{ext}"
aug_img.save(os.path.join(output_class_path, aug_name))

print("Data augmentation complete!")

# --- Training Configuration ---
BATCH_SIZE = 8

assert os.path.isdir(os.path.join(augmented_dir, "difficult")), "Missing 'difficult' folder!"
assert os.path.isdir(os.path.join(augmented_dir, "easy")), "Missing 'easy' folder!"

# --- Transforms ---
transform = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize([0.485, 0.456, 0.406],
                          [0.229, 0.224, 0.225])
])

# --- Dataset ---
dataset = datasets.ImageFolder(augmented_dir, transform=transform)
class_names = dataset.classes
print("Classes:", class_names)

# --- Train/Val Split ---
num_total = len(dataset)
num_train = int(0.8 * num_total)
indices = np.random.permutation(num_total)
```



```
train_indices = indices[:num_train]
val_indices = indices[num_train:]

train_subset = torch.utils.data.Subset(dataset, train_indices)
val_subset = torch.utils.data.Subset(dataset, val_indices)

train_loader = DataLoader(train_subset, batch_size=BATCH_SIZE, shuffle=False)
val_loader = DataLoader(val_subset, batch_size=BATCH_SIZE, shuffle=False)

# --- Feature Extractor ---
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
resnet = models.resnet18(weights=models.ResNet18_Weights.IMAGENET1K_V1)
resnet.fc = torch.nn.Identity() # remove final classification layer
resnet = resnet.to(device)
resnet.eval()

def extract_features(dataloader):
    features, labels = [], []
    with torch.no_grad():
        for imgs, lbls in tqdm(dataloader, desc="Extracting features"):
            imgs = imgs.to(device)
            feats = resnet(imgs).cpu().numpy()
            features.extend(feats)
            labels.extend(lbls.numpy())
    return np.array(features), np.array(labels)

# --- Extract features ---
print("Extracting features from training set...")
```

```
X_train, y_train = extract_features(train_loader)
print("Extracting features from validation set...")
X_val, y_val = extract_features(val_loader)

# --- Train SVM ---
print("Training SVM classifier...")
svm = SVC(kernel='linear', probability=True)
svm.fit(X_train, y_train)

# --- Evaluate ---
print("Evaluating model...")
y_pred = svm.predict(X_val)

acc = accuracy_score(y_val, y_pred)
prec = precision_score(y_val, y_pred, zero_division=0)
rec = recall_score(y_val, y_pred, zero_division=0)
f1 = f1_score(y_val, y_pred, zero_division=0)
cm = confusion_matrix(y_val, y_pred)

print(f"Accuracy: {acc:.4f}")
print(f"Precision: {prec:.4f}")
print(f"Recall: {rec:.4f}")
print(f"F1 Score: {f1:.4f}")
print("Confusion Matrix:\n", cm)

# --- Save ---
np.savez(os.path.join(augmented_dir, "svm_metrics.npz"),
         acc=acc, prec=prec, rec=rec, f1=f1)
```

```
np.savez(os.path.join(augmented_dir, "svm_confusion_matrix.npz"), cm=cm)

joblib.dump(svm, os.path.join(augmented_dir, "svm_model.pkl"))

with open(os.path.join(augmented_dir, "class_names.txt"), "w") as f:
    for c in class_names:
        f.write(c + "\n")

print("Training complete! SVM model and metrics saved.")
```

```
import os

import streamlit as st

import torch

from torchvision import models, transforms

from PIL import Image

import numpy as np


# ---- Paths & Model Setup ---

base_dir = r"D:\rupak try\airway\data"

MODEL_PATH = os.path.join(base_dir, "model_intubation.pt")

CLASS_NAMES_PATH = os.path.join(base_dir, "class_names.txt")

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")


# Local airway example images (update filenames as needed)

IMG_DIR = os.path.join(os.path.dirname(__file__), "images")

EXAMPLE_NEUTRAL = os.path.join(IMG_DIR, "6.jpg")

EXAMPLE_TONGUE = os.path.join(IMG_DIR, "5.jpg")

EXAMPLE_HEADUP = os.path.join(IMG_DIR, "4.jpg")


# ---- Streamlit App Header ----

st.set_page_config(

    page_title="AI Intubation Difficulty Predictor",

    page_icon="👤",

    layout="wide"

)


st.title("👤 Intubation Difficulty Predictor (AI-Assisted)")
```

```
st.markdown("""
AI support for preoperative and emergency airway assessment.
Upload clinical photos and get an instant risk prediction—Easy or Difficult—with
confidence scores and full transparency.
""")

# ---- About Intubation Section ----
st.subheader("What is Intubation?")
st.write("""
Intubation is a critical medical procedure to secure a patient's airway by inserting a
tube through the mouth into the trachea, ensuring proper ventilation. It is essential for:
- Major surgical procedures under general anesthesia
- Emergency situations like respiratory failure, trauma, or cardiac arrest
- Managing critically ill patients in intensive care units

Predicting difficulty in intubation is crucial for patient safety:
- Difficult intubation can lead to increased attempts, prolonged procedure time, or
complications such as hypoxia.
- Early recognition enables preparation with advanced airway tools and expert personnel.
""")
st.write("""
Difficult intubation implies:
- More attempts or prolonged intubation time
- Need for special devices such as video laryngoscopes, flexible scopes, or introducers
(bougies)
- Necessity of backup airway strategies and coordinated team efforts
""")

# ---- About the AI and Its Parameters ----
```

```

st.markdown("---")
st.subheader("How does the AI model work?")

st.write("""
- Input: Three standardized airway/face photographs per patient:
    1. Neutral position (facing camera, mouth closed)
    2. Tongue fully extended (facing camera, mouth open)
    3. Head elevated in the “sniffing” position (neck extended)
- Model: A deep convolutional neural network based on
[ResNet18](https://arxiv.org/abs/1512.03385), trained on a large dataset of labeled airway
images (easy vs difficult intubation) enhanced with data augmentation techniques:
    - RandomResizedCrop (scale 0.7–1.0)
    - RandomHorizontalFlip
    - ColorJitter
    - RandomRotation ( $\pm 15$  degrees)
    - Trained with batch size 8 for 10 epochs using Adam optimizer (learning rate 1e-4)
    - Standard mean and standard deviation normalization applied
- Output: Probability scores for “Easy” and “Difficult” intubation based on the
combined image analysis
- Purpose: To aid education, training, and clinical awareness — not to replace
comprehensive clinical airway assessment or professional judgment
""")

# ----- Example Images Section -----

with st.expander("🖼️ See Example Airway Input Photos (best practice)", expanded=False):
    coln, colt, colh = st.columns(3)

    with coln:
        st.image(EXAMPLE_NEUTRAL, caption="Neutral Face (Mouth Closed, Facing Forward)",
        use_container_width=True)

```

```
with colt:
    st.image(EXAMPLE_TONGUE, caption="Tongue Fully Extended (Mouth Open, Facing
Forward)", use_container_width=True)

with colh:
    st.image(EXAMPLE_HEADUP, caption="Head Up — Sniffing Position (Neck Extended)",
use_container_width=True)

    st.caption(
        "Capture images in good lighting with the entire face, jaw, tongue, and neck clearly
visible. Use a neutral, uncluttered medical background when possible."
    )

# ----- AI Prediction Section -----
st.markdown("---")
st.header("Upload Images for AI Prediction")

st.markdown("""
Upload the three clinical photos as described above (neutral, tongue-out, head-up). The AI
model will analyze the combined images for anatomical risk markers and provide a risk
prediction with confidence scores.

- Intended for training, skill development, and increasing airway risk awareness.
- **Not** a substitute for thorough airway examination by trained clinicians.
""")

with st.form("upload_form", clear_on_submit=False):
    form_cols = st.columns(3)

    with form_cols[0]:
        up1 = st.file_uploader("Neutral Face", type=["jpg", "jpeg", "png"], key="imgNeutral")

    with form_cols[1]:
```

```
up2 = st.file_uploader("Tongue Out", type=["jpg", "jpeg", "png"], key="imgTongue")
with form_cols[2]:
    up3 = st.file_uploader("Head Up", type=["jpg", "jpeg", "png"], key="imgHeadup")
submit_button = st.form_submit_button(label='Predict Intubation Risk')

images = []
if up1: images.append(Image.open(up1).convert('RGB'))
if up2: images.append(Image.open(up2).convert('RGB'))
if up3: images.append(Image.open(up3).convert('RGB'))

@st.cache_resource(show_spinner=False)
def load_model_and_classes():
    model = models.resnet18(weights=None)
    model.fc = torch.nn.Linear(model.fc.in_features, 2)
    model.load_state_dict(torch.load(MODEL_PATH, map_location=DEVICE))
    model.eval()
    model.to(DEVICE)
    with open(CLASS_NAMES_PATH, "r") as f:
        class_names = [line.strip().capitalize() for line in f.readlines()]
    return model, class_names

model, class_names = load_model_and_classes()

if submit_button:
    if len(images) == 3:
        imgs_resized = [img.resize((224,224)) for img in images]
        collage = Image.new('RGB', (224*3, 224))
        for i, img in enumerate(imgs_resized):
```



```

        collage.paste(img, (i*224, 0))
    st.image(collage, caption="Your Combined Image Collage", use_container_width=True)

    preprocess = transforms.Compose([
        transforms.Resize((224,224)),
        transforms.ToTensor(),
        transforms.Normalize([0.485,0.456,0.406], [0.229,0.224,0.225])
    ])
    input_tensor = preprocess(collage).unsqueeze(0).to(DEVICE)
    with torch.no_grad():
        output = model(input_tensor)
        probs = torch.softmax(output, dim=1).cpu().numpy().flatten()
        pred_idx = int(np.argmax(probs))
        st.markdown(f"""
            <div style="background:#e5fbe8;border-radius:4px;padding:1em;margin-
            bottom:0.6em;font-size:1.21em;">
                <span style="color:#16669c;font-weight:700">AI Prediction:</span>
                <span style="font-size:1.08em;font-weight:600;">{class_names[pred_idx]}
                Intubation</span>
                <br><span style="color:#289c58;font-size:1.01em;">(Confidence:
                {probs[pred_idx]*100:.2f}%)</span>
            </div>
            """, unsafe_allow_html=True)
        st.write(f"**Easy:** {probs[1]*100:.2f}% | **Difficult:** {probs[0]*100:.2f}%")
        st.caption("Interpret results in context of bedside airway exams, patient history, BMI,
        and prior anesthesia records.")
    elif 0 < len(images) < 3:
        st.warning("Please upload all 3 photos (Neutral, Tongue Out, Head Up) to get a
        prediction.")

```

# ----- Expanded Clinical Info Section -----

st.markdown("---")

st.header("Clinical Background: Difficult Intubation")

st.markdown("""

- **Difficult intubation** means that securing the airway is not straightforward and requires additional time, expertise, or specialized equipment.
- **Common predictors** include limited mouth opening, large tongue, high Mallampati score, obesity, reduced neck mobility, small jaw, airway trauma, and history of difficult intubation.
- **Clinical scores** such as Mallampati classification, thyromental distance, interincisor gap, and neck mobility tests are widely used bedside tools to assess risk.

**Why anticipate difficulty?**

- To prevent serious complications like hypoxia, trauma, and procedural delays
- To prepare the appropriate equipment and team beforehand
- To improve safety, efficiency, and confidence during airway management

This AI tool aims to support clinical training and risk awareness but should never replace comprehensive clinical assessment and decision-making.

""")

# ----- Responsible Use and Disclaimer -----

st.markdown("---")

st.markdown(

"""

<div style='border-radius:5px;padding:0.9em;background:#f2fafd;color:#155481;margin-bottom:1em'>

<b>Disclaimer:</b> This app is designed for <b>educational and simulated clinical decision support only</b>.<br>

It is **not** validated for real patient care and should never be used as the sole basis for airway management or anesthesia planning.

</div>

""",

**unsafe\_allow\_html=True**

)

## Appendix B

### Plague Report

## Rupak Ghosh

### Difficult Airway Asmnt

 Paper  
 PhD  
 Dr. A.P.J. Abdul Kalam Technical University

#### Document Details

Submission ID  
trnoid::1:3271397401

Submission Date  
Jun 8, 2025, 11:38 PM GMT+5:30

Download Date  
Jun 9, 2025, 9:59 AM GMT+5:30

File Name  
Difficult\_Airway\_Assessment\_Report.pdf

File Size  
2.4 MB

32 Pages  
5,308 Words  
31,753 Characters

Page 2 of 38 - Integrity Overview

Submission ID trnoid::1:3271397401

---

## 19% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

### Filtered from the Report

- Bibliography
- Quoted Text

---

#### Match Groups

-  **77 Not Cited or Quoted 19%**  
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**  
Matches that are still very similar to source material
-  **0 Missing Citation 0%**  
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**  
Matches with in-text citation present, but no quotation marks

#### Top Sources

- 12%  Internet sources
- 10%  Publications
- 10%  Submitted works (Student Papers)

---

### Integrity Flags

**0 Integrity Flags for Review**

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Page 2 of 38 - Integrity Overview

Submission ID trnoid::1:3271397401


Page 3 of 38 - Integrity Overview
Submission ID trnoid::1:3271397401

### Match Groups

- 
**77 Not Cited or Quoted 19%**  
Matches with neither in-text citation nor quotation marks
- 
**0 Missing Quotations 0%**  
Matches that are still very similar to source material
- 
**0 Missing Citation 0%**  
Matches that have quotation marks, but no in-text citation
- 
**0 Cited and Quoted 0%**  
Matches with in-text citation present, but no quotation marks

### Top Sources

- 12%  Internet sources
- 10%  Publications
- 10%  Submitted works (Student Papers)

---

### Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

|    |                                                                                        |  |     |
|----|----------------------------------------------------------------------------------------|--|-----|
| 1  | Student papers                                                                         |  |     |
|    | Uttar Pradesh Technical University                                                     |  | 5%  |
| 2  | Publication                                                                            |  |     |
|    | R. N. V. Jagan Mohan, B. H. V. S. Rama Krishnam Raju, V. Chandra Sekhar, T. V. K. P... |  | 1%  |
| 3  | Publication                                                                            |  |     |
|    | Bin Wang, Xiaojun Li, Jianling Xu, Bingcheng Wang et al. "Comparison of Machine ...    |  | <1% |
| 4  | Publication                                                                            |  |     |
|    | Thangaprakash Sengodan, Sanjay Misra, M Murugappan. "Advances in Electrical ...        |  | <1% |
| 5  | Student papers                                                                         |  |     |
|    | Goldsmiths' College                                                                    |  | <1% |
| 6  | Student papers                                                                         |  |     |
|    | University of Mary                                                                     |  | <1% |
| 7  | Student papers                                                                         |  |     |
|    | University of Bradford                                                                 |  | <1% |
| 8  | Publication                                                                            |  |     |
|    | Guangzhi Wang, Chenxi Li, Fudong Tang, Yangyang Wang, Su Wu, Hui Zhi, Fan Zh...        |  | <1% |
| 9  | Internet                                                                               |  |     |
|    | pmc.ncbi.nlm.nih.gov                                                                   |  | <1% |
| 10 | Internet                                                                               |  |     |
|    | www.ncbi.nlm.nih.gov                                                                   |  | <1% |


Page 3 of 38 - Integrity Overview
Submission ID trnoid::1:3271397401



Page 4 of 38 - Integrity Overview

Submission ID trnoid::1:3271397401

|    |                |                                                                                     |     |
|----|----------------|-------------------------------------------------------------------------------------|-----|
| 11 | Publication    | Xiaofan Li, Bo Peng, Yuan Yao, Guangchao Zhang, Zhuyang Xie, Muhammad Usm...        | <1% |
| 12 | Internet       | ethesis.nitrkl.ac.in                                                                | <1% |
| 13 | Internet       | www.coursehero.com                                                                  | <1% |
| 14 | Internet       | www.spiedigitallibrary.org                                                          | <1% |
| 15 | Publication    | S. Kannadhasan, R. Nagarajan, Alagar Karthick, V. Kumar Chinnaiyan. "Technolog...   | <1% |
| 16 | Student papers | University of Central Lancashire                                                    | <1% |
| 17 | Internet       | arxiv-export-lb.library.cornell.edu                                                 | <1% |
| 18 | Student papers | University of Northumbria at Newcastle                                              | <1% |
| 19 | Internet       | www.theseus.fi                                                                      | <1% |
| 20 | Publication    | Xiaofan Li, Bo Peng, Yuan Yao, Guangchao Zhang, Zhuyang Xie. "Difficult airway a... | <1% |
| 21 | Internet       | unidel.edu.ng                                                                       | <1% |
| 22 | Publication    | Feiyang Ou, Julius Suherman, Chao Zhang, Henrik Wang, Sthitie Bom, James F. Da...   | <1% |
| 23 | Internet       | diva-portal.org                                                                     | <1% |
| 24 | Internet       | www.frontiersin.org                                                                 | <1% |



Page 4 of 38 - Integrity Overview

Submission ID trnoid::1:3271397401





Page 5 of 38 - Integrity Overview

Submission ID trmold::1:3271397401

|    |                |                                                                                   |     |
|----|----------------|-----------------------------------------------------------------------------------|-----|
| 25 | Student papers | University of KwaZulu-Natal                                                       | <1% |
| 26 | Internet       | etd.caltech.edu                                                                   | <1% |
| 27 | Internet       | mafiadoc.com                                                                      | <1% |
| 28 | Internet       | repository.nirmauni.ac.in                                                         | <1% |
| 29 | Internet       | romanpub.com                                                                      | <1% |
| 30 | Internet       | link.springer.com                                                                 | <1% |
| 31 | Internet       | osuva.uwasa.fi                                                                    | <1% |
| 32 | Internet       | www.nauticalchart.gov                                                             | <1% |
| 33 | Internet       | www.slideshare.net                                                                | <1% |
| 34 | Publication    | Achraf Boumhidi, Abdessamad Benlahbib, Erik Cambria, El Habib Nfaoui. "Periodi... | <1% |
| 35 | Publication    | Lavanya Sharma, Pradeep Kumar Garg. "Deep Learning in Internet of Things for ...  | <1% |
| 36 | Student papers | Munster Technological University (MTU)                                            | <1% |
| 37 | Student papers | Trident University International                                                  | <1% |
| 38 | Internet       | braininformatics.springeropen.com                                                 | <1% |



Page 5 of 38 - Integrity Overview

Submission ID trmold::1:3271397401



Page 6 of 38 - Integrity Overview

Submission ID trm:oid::1:3271397401

|    |          |                            |     |
|----|----------|----------------------------|-----|
| 39 | Internet | core.ac.uk                 | <1% |
| 40 | Internet | odr.chalmers.se            | <1% |
| 41 | Internet | silo.tips                  | <1% |
| 42 | Internet | upcommons.upc.edu          | <1% |
| 43 | Internet | webthesis.biblio.polito.it | <1% |
| 44 | Internet | www.mdpi.com               | <1% |
| 45 | Internet | www.researchsquare.com     | <1% |



Page 6 of 38 - Integrity Overview

Submission ID trm:oid::1:3271397401

# References

- [1] Khan, Zahid Hussain. (2014). Airway Assessment: A Critical Appraisal. *Airway Management*. 15-32. 10.1007/978-3-319-08578-42.
- [2] Gunawardana, K Gunasena, P Ekanayake, U Ranwala, N. (2024). Airway Assessment Challenges and Solutions in Critical Care. 10.13140/RG.2.2.31228.55686.
- [3] Kim JH, Jung HS, Lee SE, Hou JU, Kwon YS. Improving difficult direct laryngoscopy prediction using deep learning and minimal image analysis: a single-center prospective study. *Sci Rep*. 2024 Jun 20;14(1):14209. doi: 10.1038/s41598-024-65060-x. PMID: 38902319; PMCID: PMC11190276.
- [4] De Rosa S, Bignami E, Bellini V, Battaglini D. The Future of Artificial Intelligence Using Images and Clinical Assessment for Difficult Airway Management. *Anesth Analg*. 2025 Feb 1;140(2):317-325. doi: 10.1213/ANE.0000000000006969. Epub 2025 Jan 10. PMID: 38557728; PMCID: PMC11687942.
- [5] Wang G, Li C, Tang F, Wang Y, Wu S, Zhi H, Zhang F, Wang M, Zhang J. A fully-automatic semi-supervised deep learning model for difficult airway assessment. *Heliyon*. 2023 Apr 22;9(5):e15629. doi: 10.1016/j.heliyon.2023.e15629. PMID: 37159696; PMCID: PMC10163620.

- [6] Marchis, Ioan Negrut, Matei Blebea, Cristina Crihan, Mirela Alexa, Alexandru Breazu, Caius. (2024). Trends in Preoperative Airway Assessment. *Diagnostics*. 14. 610. 10.3390/diagnostics14060610.
- [7] Wilk M, Pikiewicz W, Florczak K, Jakóbczak D. Use of Artificial Intelligence in Difficult Airway Assessment: The Current State of Knowledge. *J Clin Med*. 2025 Feb 27;14(5):1602. doi: 10.3390/jcm14051602. PMID: 40095591; PMCID: PMC11900168.
- [8] Li, Xiaofan Peng, Bo Yao, Yuan Zhang, Guangchao Xie, Zhuyang Saleem, Usman. (2025). Reliable multi-modal prototypical contrastive learning for difficult airway assessment. *Expert Systems with Applications*. 273. 126870. 10.1016/j.eswa.2025.126870.
- [9] Wu J, Yao Y, Zhang G, Li X, Peng B. Difficult Airway Assessment Based on Multi-View Metric Learning. *Bioengineering (Basel)*. 2024 Jul 11;11(7):703. doi: 10.3390/bioengineering11070703. PMID: 39061785; PMCID: PMC11274261.
- [10] Crawley, S. M., and A. J. Dalton. "Predicting the difficult airway." *Bja Education* 15, no. 5 (2015): 253-257.
- [11] Naik NB, Mathew PJ, Kundra P. Scope of artificial intelligence in airway management. *Indian J Anaesth*. 2024 Jan;68(1):105-110. doi: 10.4103/ija.ija\_1228\_23. *Epub* 2024 Jan 18. PMID : 38406331; PMCID : PMC10893795.